# Bihar Engineering University, Patna
## B.Tech. 8th Semester Examination, 2024

*Time: 03 Hours*
*Full Marks: 70*

***Course: B.Tech.***
***Code: 101818***

***Subject: Railway Engineering***

***Instructions:-***
*(i) The marks are indicated in the right-hand margin*
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all*
*(iv) Question No. 1 is compulsory*

**Q.1** ***Choose the correct answer of the following (any seven questions only):*** **[2 x 7 = 14]**

(a) The yards where trains and other loads are received, sorted out, trains formed and dispatched onwards are known as
(i) Marshalling yards
(ii) Locomotive yards
(iii) Station yards
(iv) Goods yards

(b) The composite sleeper index determines the?
(i) Suitability of the wooden sleepers
(ii) number of sleepers per rail length
(iii) permissible stresses in the steel sleepers
(iv) iv. all of these

(c) On Indian railway, the number of fish bolts used to connect one pair of fish plates to the rails are?
(i) 2
(ii) 4
(iii) 6
(iv) 8

(d) What is the objective of coning of wheels?
(i) to save material
(ii) For easy manufacturing of wheels
(iii) For easy movement on curves
(iv) For aesthetic look

(e) What is the distance between the gauge faces of tongue rail and stock rail called?
(i) Throw of switch
(ii) Heel divergence
(iii) Switch distance
(iv) Flange way

(f) Which sleepers require insulating pads for track circuiting?
(i) Wooden sleepers
(ii) Steel and CI sleepers
(iii) Concrete sleepers
(iv) Steel sleepers

(g) Which gradient determines the maximum load that can be hauled by a locomotive on that section?
(i) Rising gradient
(ii) Falling gradient
(iii) Ruling gradient
(iv) Momentum gradient

(h) A per the prescribed value, what should be the maximum value of cant excess on Broad gauge?
(i) 65mm
(ii) 75 mm
(iii) 92 mm
(iv) 100 mm

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(i) Obligatory points are also known as
(i) Controlling points
(ii) Saddle points
(iii) Rolling points
(iv) Important points

(j) On which portion of the track is a Transition curve provided?
(i) Between straight line and circular curve
(ii) Before straight line
(iii) On one side of circular curve
(iv) In the middle of circular curve

Q.2 (a) A branch track $3^0$ curve is diverted from main track $2^0$ curve in opposite direction (both are BG track) Find out the maximum speed on branch track, if maximum speed permitted on main track is 90 kmph [7]

(b) Explain the concept of negative super elevation. [7]

Q.3 (a) What is mean be a crossing? Discuss various type of crossing used in Indian railways [7]

(b) A M.G track has a sleeper density of (n+5) If the track is laid with a rail of 11.89m length, Find out the number of sleepers per one rail length [7]

Q.4 (a) With neat sketch, explain what is coning of wheels? What are the advantages and disadvantages of coning of wheels? [7]

(b) Discuss the objects of Signalling. Explain about Semaphore and Warner Signals [7]

Q.5 (a) What are the functions of sleepers? What are the different types of sleepers used in the track on Indian railways? [5]

(b) Explain the major defects in rails and its remedial measures. [9]

Q.6 (a) What is the purpose of providing marshalling yards? Describe the layout of a typical Marshalling yards. [7]

(b) What should be the actual ruling gradient (i.e. compensated gradient), if the ruling gradient as 1 in 250 has been fixed on a B G section and a horizontal curve of $3^0$ is also to be introduced over it. [7]

Q.7 (a) What is necessity of geometric design of a railway track? Enumerate the significant features of design of a railway track. [7]

(b) Explain briefly the functions of the following in a railway track [7]
I. Fish plate II. Tie bar III. Screw spike IV. Hook bolt

Q.8 (a) Explain what is meant by track capacity and discuss important measures for increasing the line capacity [7]

(b) What is the ballast in permanent way? Mention the function of ballast and state the requirements of a good ballast materials. [7]

Q.9 Write Short notes on ***any two*** of the following [7 x 2=14]
(a) Gradient and grade compensation on curves.
(b) Types of rail joints
(c) Draw the Schematic diagram of left-hand turnout. Show its various component parts



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